Exercise Prescription Cervical & Thoracic

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Cervical Spine

- Acute <6 weeks, sub-acute (6-12), chronic >12 weeks
- Chronic neck pain has the potential to change the cervical region biomechanics by adversely affecting the muscle imbalance and head-neck posture.
- Forward head posture problems are commonly seen in persons with neck pain.
- Systematic Review Findings: **Low to high certainty of evidence for positive effects on pain and disability of the various exercise types used in chronic neck pain compared to non-exercise interventions. Since no specific exercise type was found to be superior, the authors suggest that patient preference and therapist expertise should guide the choice of treatment. The most important factor is that the patient engages in *some* form of active movement rather than remaining sedentary or relying solely on passive treatments (like massage or heat). (Rasmussen-Barr et al., 2023)**

Rasmussen-Barr et al., 2023

Cervical Spine

- **Buckets of activity found in the SR:**
  - Motor Control Exercises (MCE): Focus on the deep neck flexor muscles and coordination.
  - Resistance Training: Strength training for the superficial muscles of the neck and shoulder girdle.
  - Pilates: Exercises focusing on core stability, posture, and flexibility.
  - Yoga: Mind-body practice involving specific poses and breathing.
  - Tai Chi / Qigong (TCE): Gentle flow and movement-based mind-body exercise.
- **Other Key Findings:**
  - Everything is better than nothing: There is low to high certainty that MCE, Pilates, resistance training, TCE, and yoga all provide short-term relief for neck pain compared to doing no exercise at all.
  - Impact on Disability: Most exercises (except for resistance training in some instances) also showed positive short-term effects on reducing disability.
  - The "No Winner" Conclusion: When researchers compared different types of exercises against each other, they found no significant difference. No single exercise type emerged as the "gold standard" or "best" option.
  - Sustainability: Most evidence focused on short-term results (immediately after the program). Long-term data was less certain, particularly for yoga.

Rasmussen-Barr et al., 2023

Deep Cervical Neck Flexors

- The deep cervical neck flexor muscles provide postural endurance support, and it is known that activation of these muscles is impaired in people with neck pain.
- The deep cervical flexor muscles support the cervical lordosis and are an important stabiliser of the head and neck posture.
- In individuals with mechanical neck pain, there is a decrease in muscle activation of the deep cervical muscles, causing superficial muscles such as the trapezius to undertake postural functions.
- A low load program of craniocervical flexion exercise focusing especially on motor control of the deep neck flexors has been shown to reduce neck pain and headache.

Deep Cervical Neck Flexors

- Focuses on training the deep flexor muscles:
  - Longus Capitis and Longus Colli.
  - Cues: **Tuck your chin in or hold your head up**
- Can be done with an inflatable cuff pressure sensor.
- Baseline of 20mmHg.
- Gently and slowly nod the head to 2mmHg
- Continue along: 2 → 4 → 6 → 8 → 10mmHg is reached
- Hold for 10 seconds x 6 reps



Fig 1. *The clinical application of the craniocervical flexion test. The patient is guided to each progressive pressure increment of the test by feedback from the pressure sensor. The clinician analyses the movement and detects the presence of any activity in the superficial flexors.*

Jull GA, O'Leary SP, Falla DL. Clinical Assessment of the Deep Cervical Flexor Muscles: The Craniocervical Flexion Test. *Journal of Manipulative and Physiological Therapeutics.* 2008; 525-533.

1 Deep neck flexors



Sets: 2 Reps: 12

Sit with the back of your head on the wall.
You can place your hand across the front of your neck just above your collarbone to monitor unwanted activity of the surface muscles.
Place your tongue on the roof of your mouth, keeping your jaw relaxed.
Keeping the head in contact with the wall, slide the back of the head up the wall as you nod your chin down as far as you can go without the surface muscles tightening under your fingers.
Hold for the time instructed and then return to the starting position.

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2 Deep cervical flexion



Sets: 2 Reps: 12

Lie on your back with a towel under your neck.
Tuck your chin inward and get yourself tall by pretending that a rope is pulling you up from the back of your head.
Slightly lift your head from the towel, keeping the chin tucked in.
You should be able to swallow at all times during the exercise.
Hold for the prescribed time and repeat.

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3 Repeated cervical retraction



Sets: 2 Reps: 12

Stand or sit upright.
Look at an object at eye level and slowly move your head back, tucking your chin.
Guide the movement by placing your finger on your chin.
Return to the starting position and repeat.

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4 Chin tuck w/ elastic



Sets: 2 Reps: 12

Place a rubber band on the back of your head.
Hold the rubber band with your elbows bent and your hands in front of your face.
Keep your back straight.
Tuck your chin in without tilting your head, until you get a double chin.
Pull the elastic in front of you and hold.

Considerations:

- It is important to consider what is occurring above and below the joint
  - E.G. neck - shoulder, scapular, thoracic spine, upper body (bicep, tricep)
- **Using specific strengthening exercises as a part of routine practice for chronic neck pain, cervicogenic headache and radiculopathy may be beneficial. Strengthening and endurance exercises for the cervico-scapulothoracic and shoulder may be beneficial in reducing pain and improving function.**
- THINK:

1. What stage of healing? Acute, subacute, chronic
2. What is the impairment? ROM, PAIN?
3. What type of exercise is suitable? ROM, Isometric, Isotonic, Mobility?
4. How can I get this person engaging in rehab?

1 Left head translation



Sets: 2 Reps: 12

Put your left shoulder against the wall to block the trunk movements
Put a band around the head to apply some resistance to the exercise.
Keep your face facing forward and perform a translation of the head toward the left, against the resistance of the band. Then come back to the center.
You should not incline or rotate the head while performing the movement.

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2 Stabilization flex./ext.



Sets: 2 Reps: 12

Stand in a split stance and tie an elastic in front of you at waist level.
Firmly hold each end of the elastic with each hand.
With your chin tucked in, pull the tip of your shoulder backwards and place your hands and arms in line at shoulder level with your palms facing down.
Move one arm over your head, keeping it straight, while the other moves close to the body downwards simultaneously.
Slowly return to the starting position, keeping your chin tucked in and repeat going the opposite way.

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3 Row rotation, elastic



Sets: 2 Reps: 12

Stand in a split stance.
Tie an elastic in front of you at waist level.
Hold the elastic tightly with one hand. Your back leg is on the same side as the arm pulling.
Your elbow is flexed to 90 degrees.
Pull the elastic backwards as far as possible by joining your shoulder blades together and rotating your upper body. Your head and hips must remain facing straight.
Return slowly to the initial position and repeat.

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4 Face pull with band



Sets: 2 Reps: 12

Hold a band that is anchored in front of you. With the palms facing down, pull the band toward your neck with the elbows high and the shoulder low. Keep your back straight and don't poke the chin forward as you pull.

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5 Scapula retraction



Sets: 2 Reps: 12

Lie on your stomach, face down with your arms overhead, elbows bent and thumbs pointing upward.
Lift your elbows and hands off the surface while pulling your shoulder blades together and downward.
Let your elbows down and repeat.

Cervical ROM:

- Flexion
- Extension
- Side flexion
- Rotation

1. Can perform active ROM to improve range and tolerance
2. Can perform isometric training to relieve neck pain, improve neuromuscular control and joint loading.

1 Active neck extension



Sets: 2 Reps: 12

Stand or sit tall.
Move your head backwards as to look up to the ceiling.
Return to the initial position and repeat.
Do not tilt or turn your head.

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2 Neck flexion/extension



Sets: 2 Reps: 12

Start by assuming a nice and tall posture with the chin slightly tucked in.
Slowly look down and gently bring your chin down towards your chest.
Slowly roll your head and neck backwards to look up at the ceiling as far as is comfortable.
Repeat the process, keeping the chin tucked in.

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3 Cervical extensor exercise



Sets: 2 Reps: 12

Start on all fours with a neutral neck position and your chin slightly tucked.
Lower your head segmentally to bring the chin to your chest. Think about rolling the neck vertebrae by vertebrae, starting with the head.
Reverse the steps to lift the head, starting with the base of the neck and finishing with the head.

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4 Stretching side bending



Sets: 2 Reps: 12

Lift one arm and bring it up and across your head.
Sit straight and place the palm of your hand on your head.
Use your hand to pull your head gently down towards your shoulder until a stretch is felt on the opposite side of the neck.
Maintain the position.
Do not turn your head or tilt it forward or backwards while you perform this stretch.

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5 Active neck rotation



Sets: 2 Reps: 12

Position yourself in a tall sitting or semi-reclined position.
Turn your head to one side for the prescribed number of repetitions.
Repeat on the other side.

Cervical Spine: Loading

1. Dumbbell Shrug
2. One arm row
3. Upright row
4. Reverse Fly
5. Lateral Raises
6. Lat pulldown
7. Face Pull
8. Front raise
9. Plank (high / elbow)

6 Shoulder flex, supination



Sets: 2 Reps: 12

Step on one end of the elastic and hold the other end in your hand.
Tuck your chin in, pull the tip of your shoulder backwards and turn the palm of your hand facing forward before you raise your arm in front of you, keeping your elbow straight and your head still.
Lower your arm slowly and repeat.

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7 Shoulder extension w/ elastic



Sets: 2 Reps: 12

Stand in a good upright posture with knees slightly bent and hold this posture throughout the exercise. Do not let the chin poke.
Hold the ends of the elastic with your elbows straight and your palms facing down.
Initiate the motion with the shoulder blades, opening up through the front of the shoulders and drawing the blades backward. Do not shrug.
Then slowly pull the arms down to your sides-not behind you.
Pause at the end of the movement, then slowly return to the starting position.
Concentrate on quality of the movement, controlling your head and shoulders posture before increasing the resistance.

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8 Scapula retraction



Sets: 2 Reps: 12

Lie on your stomach, face down with your arms overhead, elbows bent and thumbs pointing upward.
Lift your elbows and hands off the surface while pulling your shoulder blades together.
Let your elbows down and repeat.

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9 Shrugs with weight



Sets: 2 Reps: 12

Stand and hold a weight in each hand.
Raise your arms apart from your body to 30 degrees and pull the tip of your shoulder backward while you lift your shoulders towards your ears, keeping your chin tucked in.
Slowly lower your shoulders and repeat.

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10 Plank with band



Sets: 3 Reps: 20-30 secs

Place a taut band around your wrists, with arms shoulder width apart, and forearms and hands on the ground with thumbs up.
Push up into a plank position keeping your head in a neutral position.
Hold the position for the prescribed time-or until unable to hold the position correctly.
Moving the band closer to the elbows will decrease the resistance and make the exercise easier.

Thoracic ROM:

- Mobility
- Cat Camel
- Thread the needle
- Open book
- Thoracic rotations
- Foam Roller
- Thoracic extension
- T-spine self mobilisation
- Thoracic extension four point

1 Thoracic rotation



Sets: 2 Reps: 12

Lie on your side with the hips and knees bent in front of you.
Both hands are placed together in front of your chest.
Keep the bottom arm on the floor, roll back and have the top arm reach back toward the floor, opening up the chest. Make sure you keep your arm in line with the collar bone to preserve the shoulder's integrity.
Think about bringing your shoulder blade down to the floor.
Maintain the position and return slowly to the starting position.

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2 Arm openers



Sets: 2 Reps: 12

Lie on your side with your hips and knees flexed, head supported and arms extended in front of you with your hands together.
Protract your shoulder blade by reaching forward with your top hand, then retract back and slowly lift your arm and rotate your trunk to reach the ground on the opposite side.
Make sure you look at your fingertips during the entire movement to make sure you are rotating your spine and not only moving your arm.
Come back to the starting position and repeat.
Your lower back, hips and legs should remain still during the exercise.

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3 Thread the needle



Sets: 2 Reps: 12

Start in a four point position with your hands directly under the shoulders and the knees under the hips.
Keep your chin in, back straight and shoulders back.
With one arm reach under the other as far as possible rotating and rounding your upper back.
Bring your arm back and reach back in the opposite direction as far as possible.
Keep your elbow bent to 90 degrees and look at your moving hand at all times.
Repeat with the other arm if indicated.

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4 Thoracic extension mobility



Sets: 2 Reps: 12

Place your foam roller on the floor and lie on your back with your knees bent and the foam roller at the level of your shoulder blades.
With your arms behind your head, lower your head as close as you can to the floor until you feel a stretch behind your back.
Maintain your abs tight and proper low back posture during the exercise.

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5 Thoracic ext. mobility



Sets: 2 Reps: 12

Place your foam roller on the floor and take a kneeling four point position with your wrists on the foam roll in front of you.
Roll the foam roll forward by stretching your spine and lowering your chest towards the floor.

6 T-spine self-mobilization



Sets: 2 Reps: 12

Position the foam roller perpendicular to your body and place the top of your shoulder blades on the foam roller.
Support your head with your hands and bring your elbows together as much as possible.
Lift your hips so your back is straight.
Roll the foam roller down your spine-middle back-by using your legs.
Roll the foam roller back to the starting position-at your shoulders.
You may perform the specified number of repetitions consecutively or give your legs a break by bringing your hips down to the ground for a brief rest.

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7 Upper spine/neck release



Lie down on your back on a foam roller.
First, put a double tennis ball taped together in your back-at the level of the shoulder blades.
Release the area by rocking side to side or by moving the arms.
Second, put a folded towel underneath the neck and the tennis balls over it. Relax the neck.
You can move back and forth to accentuate the release.
Third, put the towel under the neck and the tennis balls at the base of the skull. Perform gentle side-to-side and up-and-down movements.

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8 Foam roll thoracic spine



Lie over a foam roller positioned at shoulder blade level.
Cross your arms over your chest and roll your body on the foam roller from the middle of your back to the base of your neck.
Do not go too fast when you execute the movement. You can focus on tender spots.
Breathe normally.

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9 Thread the needle with roller



Sets: 2 Reps: 12

Start in a four point position with the hands directly under the shoulders and the knees under the hips.
Keep your chin tucked in, back straight and shoulders back.
With one arm, reach under the other onto the roller, palm up, and roll the arm and shoulder through.
Return to the center and raise the arm up and back to open your chest.
Repeat with the other arm if indicated.

Thoracic: Loading

- Bird-Dog
- Single arm row
- Bent over row
- Thoracic rotations/ lunge
- Push-ups
- T, Y, I
- Thoracic Extension
- Face pull
- Bridge
- Lat pulldown
- Squat / squat rotations
- Plank (normal or side plank)

1 Bird dog



Sets: 2 Reps: 12

Get on your hands and knees (four point position) with your knees directly under your hips and your hands directly under your shoulders.
Your back is in a neutral position (slightly arched) and your chin must be tucked in.
Tighten slightly your abdominals, lumbar muscles and pelvic floor muscles then lift one arm and the opposite leg without allowing the trunk or pelvis to move or rotate.
Try to grab something far away in front of you with your hand and touch an imaginary wall far behind you with your foot instead of just lifting them up.
Lower your leg and arm back to the floor and repeat with the other leg and the opposite arm.

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2 Push-up



Sets: 2 Reps: 12

Lie on your stomach with your hands beside your shoulders.
Raise your body off the floor by straightening your elbows, keeping your chin tucked in and your body straight as a plank.
Bend your elbows and lower yourself back down to the floor.
Repeat.

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3 Trunk rotation, elastic



Sets: 2 Reps: 12

Stand in a lateral split stand.
Tie an elastic high up to one side of a door.
Hold the elastic tightly with both hands at shoulder level with your shoulders back and your weight on the leg closest to the door.
Transfer your weight to the other leg, rotate your trunk using your abdominals and upper back and pull the elastic towards the floor.
Slowly return to the initial position with control and repeat.

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4 One leg row, elastic



Sets: 2 Reps: 12

Tie an elastic in front of you at waist level.
Hold the elastic tightly in one hand.
Lift one leg up (elastic side) and pull the elastic backwards as far as possible by joining your shoulder blades together and moving your arm back. Maintain your balance on one leg with your shoulders back and your trunk stable during the exercise.
Slowly return to the initial position and repeat.

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5 Strengthening extension



Sets: 2 Reps: 12

Lie face down on a mat with your arms behind your head and your chin tucked in.
Lift your upper body up by pulling your shoulder blades together without lifting your feet.
Return to the starting position and repeat.

6 Face pull with theraband



Sets: 2 Reps: 12

Hold the ends of a theraband anchored to an object located in front of you. With the palms facing down, pull the band toward your neck with the elbows high and the shoulders low. Keep your back straight and don't poke the chin forward as you pull.

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7 Floor block (3 positions: I-Y-T)



Sets: 2 Reps: 12

Lie face down with your wrists on a 6-inch block, or a similar elevated object, elbows straight and inward pointing toes. Make list fists with the hands (don't clench them tightly) and point the thumbs upward. Rotate your arms from the shoulders rather than the elbows. Your forehead should be on the floor with your neck, shoulders, buttocks and stomach relaxed. Let your hips fall into the ground.
Hold each position for a minute-or for the recommended time:
1) Arms over head forming an "I";
2) Arms in a 45 ° angle forming a "Y";
3) Arms on the side at a 90 ° angle forming a "T".

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8 Rotation lunge with band



Sets: 2 Reps: 12

Tie an elastic beside you at waist level and hold the other end in your hands.
Get into a lunge position, with both knees at 90 degrees without touching the ground with your back knee.
Rotate your trunk to the opposite side of the elastic without turning your pelvis.
Return to the middle and repeat.

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